

Lecture 3

# The Hopfield Network Pt.2

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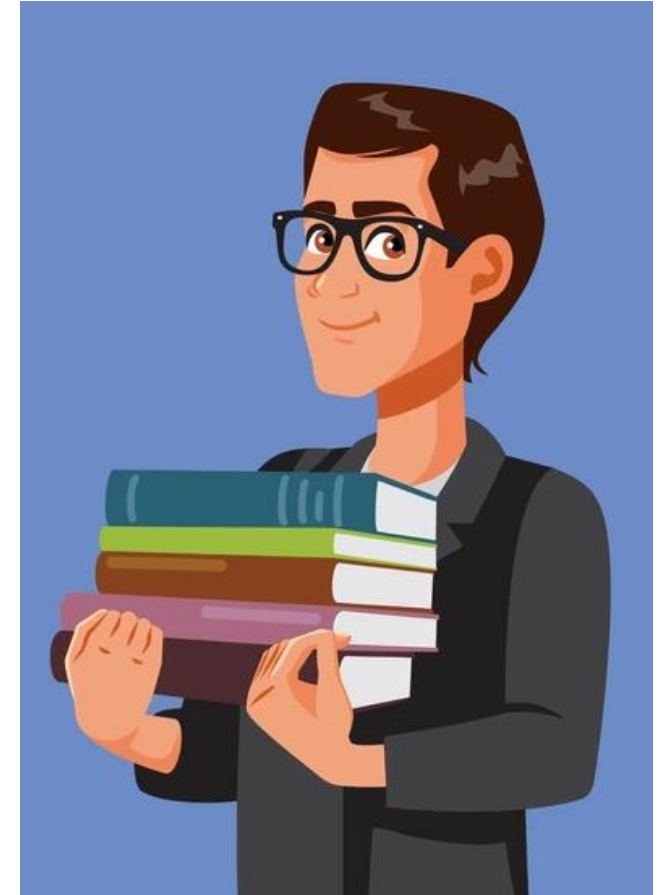
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# Why the Hopfield Network?

1. A **Hopfield Network (HN)** is a recurrent neural network capable of storing and retrieving **multiple** memories
2. It addresses the **Associative Memory Problem**
3. Single Stored Memory  
vs Multiple Stored Memories



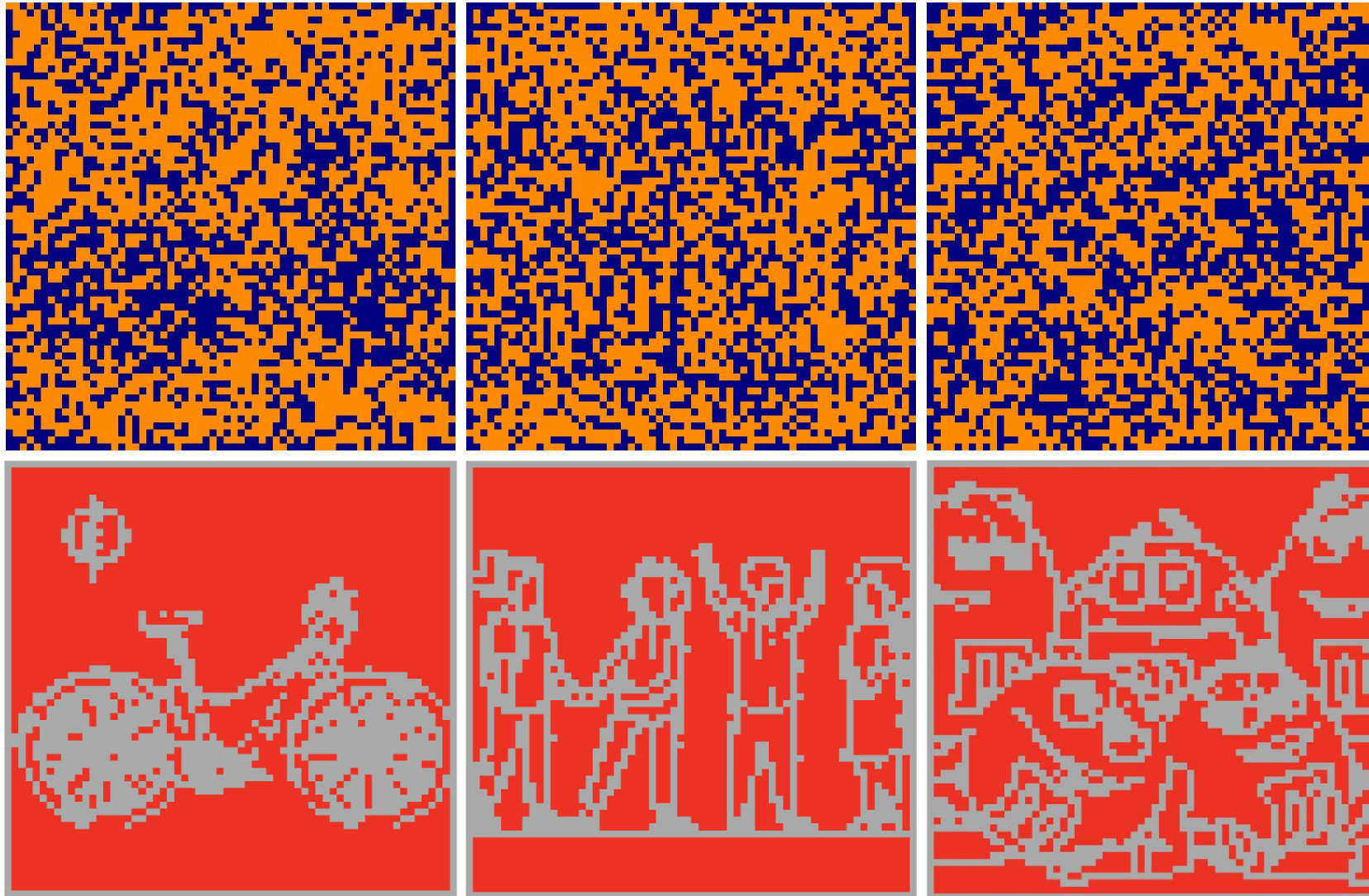
# Multiple Stored Patterns

1. How do we get the system to restore the most similar pattern to the test one out of many patterns:  $\vec{\xi}^1, \vec{\xi}^2, \dots, \vec{\xi}^p$ ?
2. We define the weights as a **SUPERPOSITION OF TERMS**:

$$w_{ij} = \frac{1}{N} \xi_i \xi_j \quad \longrightarrow \quad w_{ij} = \frac{1}{N} \sum_{\mu=1}^P \xi_i^{\mu} \xi_j^{\mu}$$

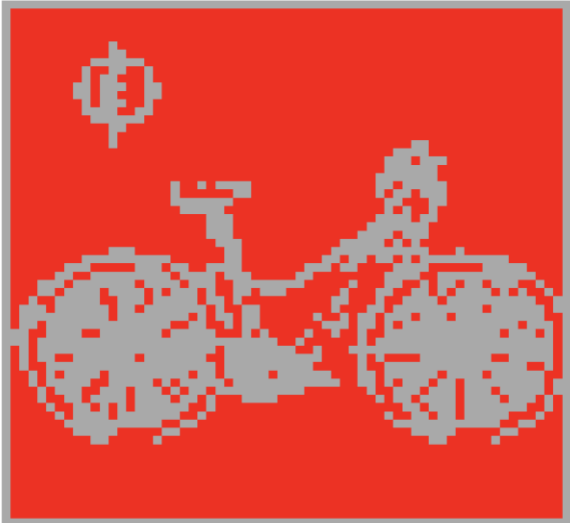
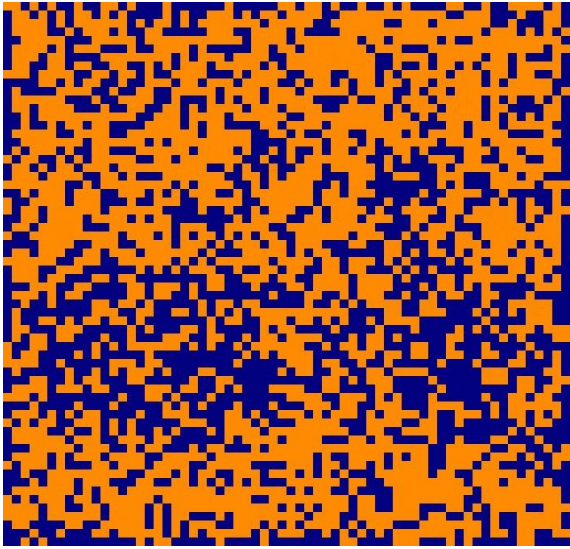
1. Multiple stored patterns:
  - Enable associative memory: Partial inputs can retrieve full patterns.
  - Make efficient use of the network's capacity.
  - Support pattern recognition applications.

# Example of multiple stored patterns



[Source: <https://towardsdatascience.com/hopfield-networks-neural-memory-machines-4c94be821073>]

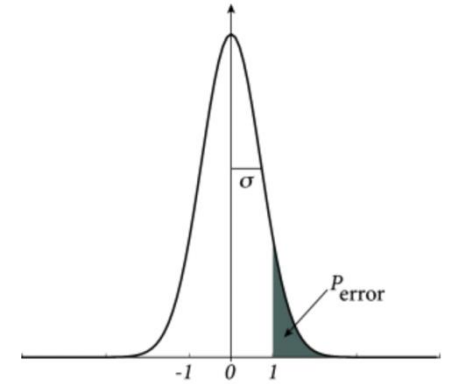
# Example of multiple stored patterns



- In this example above, a HN of 4096 neurons has memorized three distinct patterns (the red images)
- Each pattern is a 64x64 pixel image
- The network is initialized 3 times from a distorted version of each image
- The network restores each respective pattern as its state converges
- **CAPACITY:** Networks configured in this way are actually capable of storing up to  $\sim 0.15N$  patterns, where  $N$  is the number of neurons in the network (we will see this in more detail)

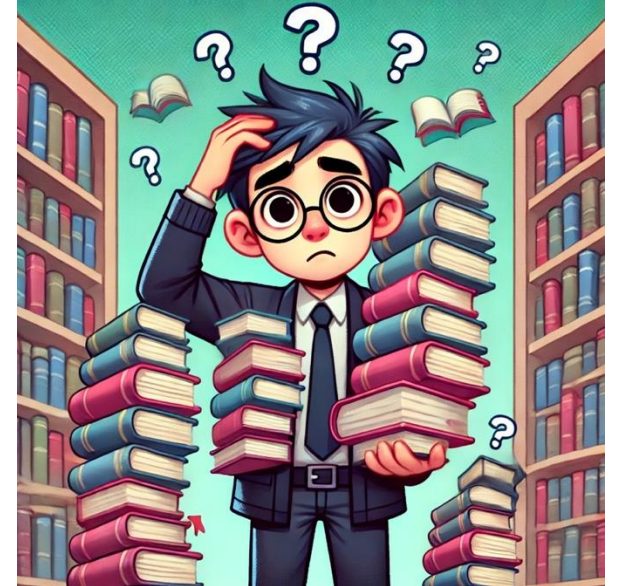
# Storage Capacity

1. The capacity of a Hopfield Network (HN) refers to the **maximum number of patterns it can store and accurately retrieve** ( $P_{\max}$ ).
2. The network stores patterns as stable attractors in an energy landscape.
3. If too many patterns are stored, retrieval becomes unreliable due to **interference** and **spurious states**.



## Ways to Improve Capacity:

1. **Sparse coding:** store patterns where only a fraction of neurons activate per pattern.
2. **Modified learning rules:** use Storkey Learning Rule or other variations to improve separation between patterns.
3. **More Neurons:** increasing  $N$  directly increases  $P_{\max}$ .
4. **Hierarchical Networks:** structuring memory storage can reduce interference.



# Energy Function

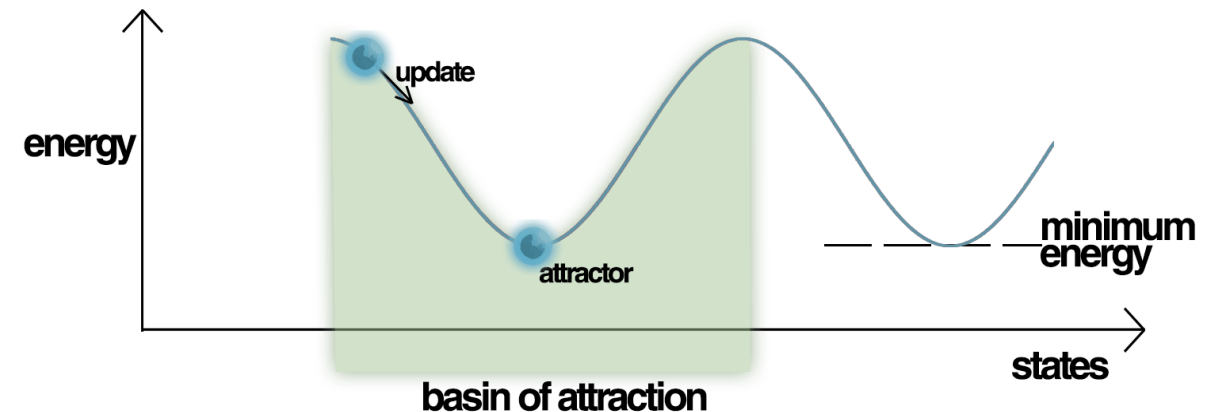
- The model:

$$O_i(t + 1) = g \left( \sum_j w_{ij} O_j(t) + b_i \right)$$

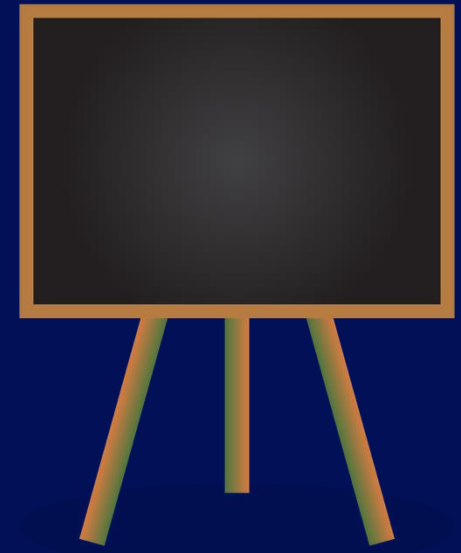
- **Stable states** are the minimizer of the energy function:
- The **Energy function** monotonically decreases with each subsequent update generation
- We could start from this formula and use automatic gradient descent algorithm to find the weights!

Energy Function

$$E(\vec{x}) = -\frac{1}{2} \vec{x}^\top \mathbf{W} \vec{x} - \vec{x}^\top \vec{b}$$



# Let's formalize on the blackboard



# Want to Learn More?

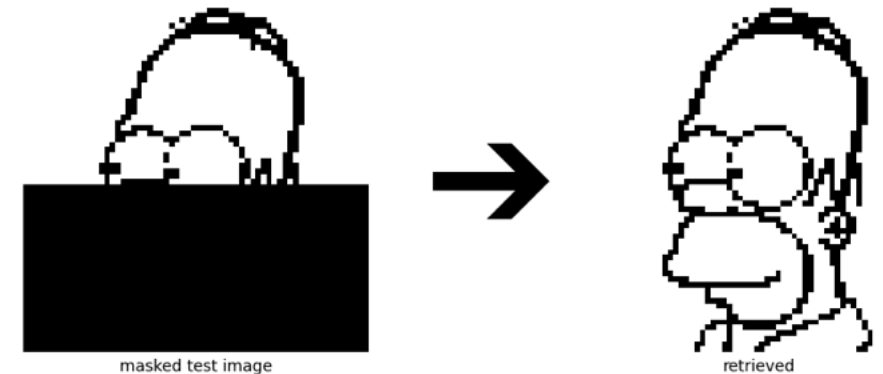
- Suggested additional reading for enthusiasts (not asked in the written exam):

Ramsauer, H., Schäfl, B., Lehner, J., Seidl, P., Widrich, M., Adler, T., Gruber, L., Holzleitner, M., Pavlović, M., Sandve, G. K., Greiff, V., Kreil, D., Kopp, M., Klambauer, G., Brandstetter, J., & Hochreiter, S. (2020). Hopfield networks is all you need. arXiv preprint arXiv:2008.02217.

<https://arxiv.org/abs/2008.02217>

- More accessible explanation in the following blog post:

<https://ml-jku.github.io/hopfield-layers/>





# Thank you – Any questions?



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